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Optical coherence tomography in the assessment of suspicious oral lesions: An immediate *ex vivo* study

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Epithelial thickness;
Sensitivity;
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Summary
Background: Optical coherence tomography (OCT) is an evolving optical technology that is capable of delivering real-time, high-resolution signatures of tissue.
Objectives: The purpose of this immediate *ex vivo* prospective clinical study was: (1) to assess the sensitivity and specificity of OCT on biopsy material in identifying potentially malignant and malignant oral lesions, (2) to determine the inter-observer agreement in the analysis of specific image parameters, and (3) to find out the oral epithelial thickness for different pathology groups.
Materials and methods: This prospective study involved 125 suspicious oral lesions from 125 patients. The lesions were surgically biopsied and subjected to OCT in the immediate *ex vivo* phase. Two independent readers (surgeon and pathologist) examined the OCT images and assessed several cellular features including keratin layer, epithelial layer, basement membrane and lamina propria, and recorded their findings using special OCT reading score. The sensitivity, specificity and accuracy of OCT to predict ''the future need for surgical biopsy in case of any similar lesion'' were calculated. The epithelial thickness was also measured. The degree of agreement between the two readers was recorded.
Results: The pathological diagnosis revealed that the majority of lesions demonstrated microinvasive carcinomas ($n = 43$). Forty-one had different degree of dysplasia. Benign oral lesions were less common and included 22 keratosis, 11 non-specific lesions, 6 mucocels and 2 papillomas. Optical coherence tomography achieved a sensitivity of 85% and a specificity of 78% in the assessment of oral potentially malignant and malignant disorders. The positive and negative predictive values were 86.5% and 77.5%, respectively. The accuracy was 82% and the kappa

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coefficient of inter-observer agreement was 0.72 on ''the need for biopsy''. OCT imaging of oral lesions provided valuable information on the oral epithelial thickness.
Conclusion: This study proposes that OCT can accurately identify wide spectrum of oral tissue pathologies. Further studies can assess the role of OCT in evaluating and guiding surgical biopsies and monitoring disease.

Introduction

Oral cancer represents one of the most challenging diseases in head and neck. Despite the advancement in cancer targeting therapy, survival rates post intervention has remained the same over the last 50 years [1]. Any apparent gain can be attributed to lead time bias of detection systems with no apparent increase in real survival. Most of the published data around the world seems to agree on a figure just above 50% to be the 5-year survival of oral squamous carcinoma patients. One can attribute this to the late presentation of patients and/or lack of early detection mechanisms.

The process of oral neoplasia (carcinoma) can be preceded by early dysplasia (*i.e.* potentially malignant disorder); where changes occur on the cellular and subcellular level but with no breach of the basement membrane occurs (a defining characteristic of malignancy). Early detection and intervention in these potentially malignant disorders can prevent progression. Hence, the appropriate treatment of these pre-malignant disorders along with elimination of risk factors (*i.e.* smoking, drinking, betel nut chewing and human papilloma virus immunization) can theoretically prevent the development of cancer.

Eighty-five percent of oral potentially malignant disorders and/or malignancies may present as visibly detected white lesions (leukoplakias), with variable degree of cellular changes. Other less commonly presentations includes red lesions (erythroplakia) and mixed lesions (leukoerythroplakia). The pathological background of these lesions is unpredictable [2].

Early detection of these potentially malignant and malignant disorders can be very advantageous; early intervention (*i.e.* medical and/or surgical) can reduce morbidity and mortality. A patient, classically, presents to the general dental or medical practitioner complaining of painless ulcer on the lateral border of tongue or floor of mouth. Visual and clinical assessment of the troublesome lesion is essential and can help to determine the need for surgical biopsy to confirm the suspicion of potentially malignant or malignant disorder. This remains an area of subjective variation with the potential loss of treatment time and worse outcomes.

Several authorities have identified general dental practitioners as being highly sensitive with low specificity when it comes to identifying these conditions; while general medical practitioners (with less training in identifying oral pathologies) are less sensitive and more specific. Furthermore, differentiating between oral lesions which share similar macroscopic clinical characteristics (*i.e.* dysplasia and carcinoma *in situ*) can be sometimes difficult even for specialists in the area [3].

Thus an aid is required for the general practitioner and the specialist. This can help in identifying suspicious lesions requiring a surgical biopsy, monitoring of potentially malignant disease processes not requiring intervention as well as macroscopically healthy areas adjacent to a previously excised disease process. Further applications include guiding biopsies and assessment of surgical margins.

Histopathological analysis of excision biopsy specimens continues to be the gold standard diagnostic procedure for any suspicious lesion. However, reporting can take several days because it involves necessary tissue processing, fixation and examination of the paraffin wax embedded slides [4,5]. Exfoliative cytology and toluidine blue staining can be used to improve the clinician's ability to distinguish various suspicious lesions, but both techniques rely on subjective evaluation which can differ per clinician. False positive and false negative results have been reported with these techniques [6].

The concept of non-invasive real time optical imaging systems has become a pressing need especially with devolving care to the community. The term optical biopsy encompasses a wide range of dental and medical screening systems. To date only fluorescence spectroscopy systems are being used in clinical practice. Several others are still in clinical trials, including elastic scattering spectroscopy (ESS), differential path-length spectroscopy (DPS), near infrared spectroscopy, Raman spectroscopy, confocal imaging, microendoscopy, optical coherence tomography (OCT) and molecular imaging.

Optical coherence tomography (OCT) has significantly evolved after the invention of the interferometer. OCT, an optical modality, provides high resolution (10—20 μm) cross-sectional images of tissue *in situ*, up to 10 times higher than conventional ultrasound, magnetic resonance imaging (MRI), or computed tomography (CT) [22]. The clinical applications of OCT have been expanded to involve wide range of clinical domains. Initially, success was achieved with retinal pathology [7] as well as bronchopulmonary diseases [8]. Further encouraging results were achieved with pathologies of the skin [9], gastrointestinal tract [10], genitourinary tract [11], breast tissue [12], as well as diseases of the oropharyngeal/laryngeal region [13,14].

This prospective clinical study evaluates the use of optical coherence tomography in the assessment of biopsies from suspicious lesions of the oral cavity.

Materials and methods

Identical protocols were used to recruit 125 consecutive patients who presented with suspicious oral lesions to the UCLH Head and Neck Centre, London between 2008 and

Table 1 Demographics of the cohort included in this study.

	No. (%)		No. (%)
Gender		Medical history	
Male	72 (57.6)	ASA I	123 (98.4)
Female	53 (42.4)	ASA II	2 (1.6)
Location		Symptoms	
Post. dorsal tongue	28 (22.4)	Soreness	17 (13.6)
Buccal mucosa	20 (16.0)	Pain	16 (12.8)
Ant. dorsal tongue	12 (9.6)	Itchiness	2 (1.6)
Floor of mouth	11 (8.8)	Bleeding	1 (0.8)
Soft palate	11 (8.8)	Asymptomatic	89 (71.2)
Vermilion border	10 (8.0)	Smoking status	
Gingival mucosa	9 (7.2)	Current smoking	44 (35.2)
Ventral tongue	7 (5.6)	Ex-smoker	48 (38.4)
Lower lip	5 (4.0)	Non-smoker	33 (26.4)
Hard palate	4 (3.2)	Drinking status	
Retromolar trigone	3 (2.4)	Current drinker	89 (71.2)
Upper lip	3 (2.4)	Ex-drinker	16 (12.8)
Tonsil	2 (1.6)	Non-drinker	20 (16.0)
Colour		Pan chewing	3 (2.4)
Leukoplakia	85 (68.0)	Diagnosis	
Erythroplakia	26 (20.8)	Microinvasive carcinoma	43 (34.4)
Speckled leukoplakia	13 (10.4)	Dyspalsia	41 (32.8)
Bluish	1 (0.8)	Keratosis	22 (17.6)
Clinical features		Non-specific inflammatory	11 (8.8)
Macule	65 (52.0)	Mucocels	6 (4.8)
Papule	40 (32.0)	Papillomas	2 (1.6)
Ulcer	20 (16.0)		
Biopsy type			
Excisional	86 (68.8)		
Incisional	39 (31.2)		

2010. The study protocol was approved by the Moorfields and Whittington Local Research Ethics Committee.

All the patients presented with clinical features suggestive of suspicious oral mucosal disease. These include non-healing ulcerative lesions after removal of the causative factor. Long standing leukoplakia, erythroplakia and erythrolekoplakia are categorized as suspicious lesions. Any non-healing lesions in patient having history of oral cancer were also included in the study. Lesions smaller than 1 cm in diameter were excluded from the study.

Written informed counselled consent was obtained from every patient. Demographic information of each patient was collected in pre-made proformas. Detailed clinical examination was performed on each patient to assess the site, size, and the clinical characteristics of the lesion (Table 1). The presenting complaints, smoking and drinking habits were also highlighted.

Following examination, surgical (excisional or incisional, when appropriate) biopsies were acquired from each patient. Tissue in excess of clinical need was examined; examination did not preclude later histological analysis. Photography was used to ensure accurate co-registration of specimens subject to OCT and histological analysis to allow precise comparisons. The total number of surgical biopsies was 125. The biopsy samples were preserved in normal saline and within 1 hour (immediate *ex vivo*) were subjected to optical coherence tomography scanning. All specimens were then processed for histopathological diagnosis. Histopathological assessment was carried out by one specialist oral and maxillofacial pathologist to ensure objectivity. Assessment was conducted according to the World Health Organization guidelines. The pathological categories included: normal/benign, dysplasia (mild, moderate, severe), carcinoma *in situ*, or invasive cancer.

The process of co-localization was described in a previous study published by our group [19]. The co-registration process involved diagrams, digital images and specimen orientation using sutures and special ink. For each surgical specimen, OCT images were acquired from several areas of interest (normal and suspicious looking areas) to allow comparison.

Assessment of OCT images was carried out by a surgeon and a pathologist; both were trained to interpret OCT images and provided with training set to consolidate their knowledge and assessment skills prior to the assessment of the 125 OCT images (Figs. 1—3).

The principal investigator (study coordinator) selected the most appropriate OCT scans for the normal and abnormal part of the biopsy. This was confirmed after histopathological examination. The assessors were blinded to each other and to the patient. Each assessor was provided with brief clinical history of the suspicious lesion and was asked to

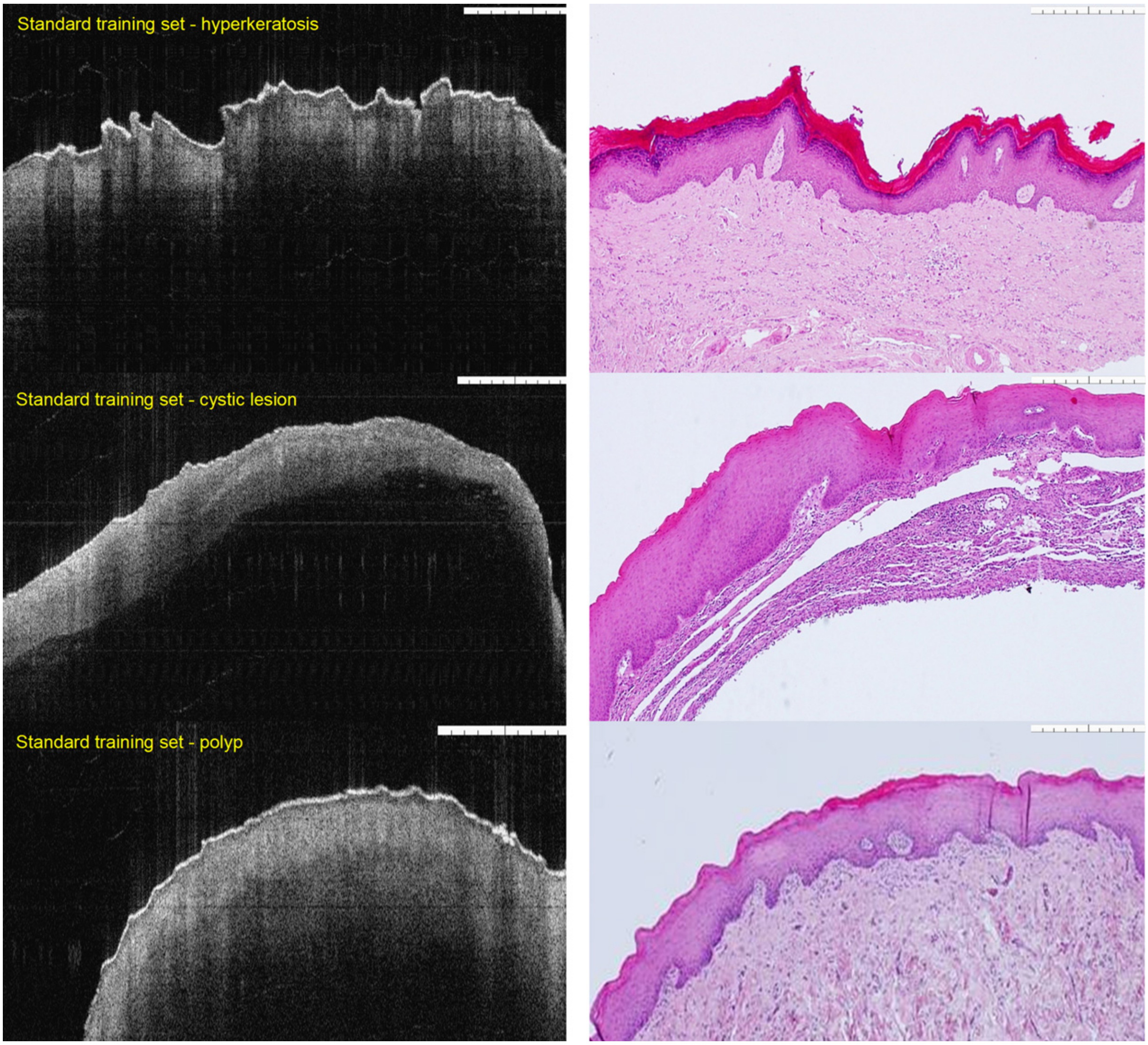

Figure 1 Data (OCT images) used in training both the clinician and the pathologist in identifying pathological processes. The images were compared to the gold standard histopathology. Here the images include hyperkeratosis, cystic lesion and polyp.

comment on every corresponding OCT image using a pre-made proforma. Metric readings were recorded for both the keratin and the epithelial layers.

Four main parameters were assessed in every image, including keratin layer, epithelial layer, lamina propria and basement membrane. The assessors were instructed to quantify the thickness (no change, increased or decreased) for each tissue (keratin and epithelium) layers as well as commenting on the status of the basement membrane (intact, breach, difficult to assess), basement membrane quality (excellent, good, adequate or poor) and lamina proprial changes (obvious changes, no clear changes or no changes). Furthermore, each assessor was required to provide a differential diagnosis and was asked to report on ''the need for surgical biopsy'' to confirm the diagnosis if this technique was to be applied *in vivo*.

In this study, we used a swept-source frequency-domain optical coherence tomography microscope (Michelson Diagnostics EX1301 OCT Microscope V1.0), the components of which are illustrated in previous study by Jerjes et al. [19]. The light source used is a Santec HSL-2000, with an imaging wavelength of 1310 nm, axial optical resolution of <10 μm, and lateral optical resolution of <10 μm. The system provides an image resolution of 5.3 μm/pixel with a maximum image width of 6 mm, a sub-surface imaging depth of 1.5 mm, and a focal depth of 1 mm. Samples can be manipulated to see the full quality results on the screen instantly, with an image capture time of <100 ms and refresh rate of >1 Hz. Images were acquired in two modes: black and white, and inverted colour mode.

Statistical analysis

Working upon the data given by both assessors, the sensitivity, specificity, and overall accuracy of OCT were calculated. Agreement was assessed by using Kappa scores (poor = 0.00—0.40, good = 0.41—0.70, very good = 0.71—0.80,

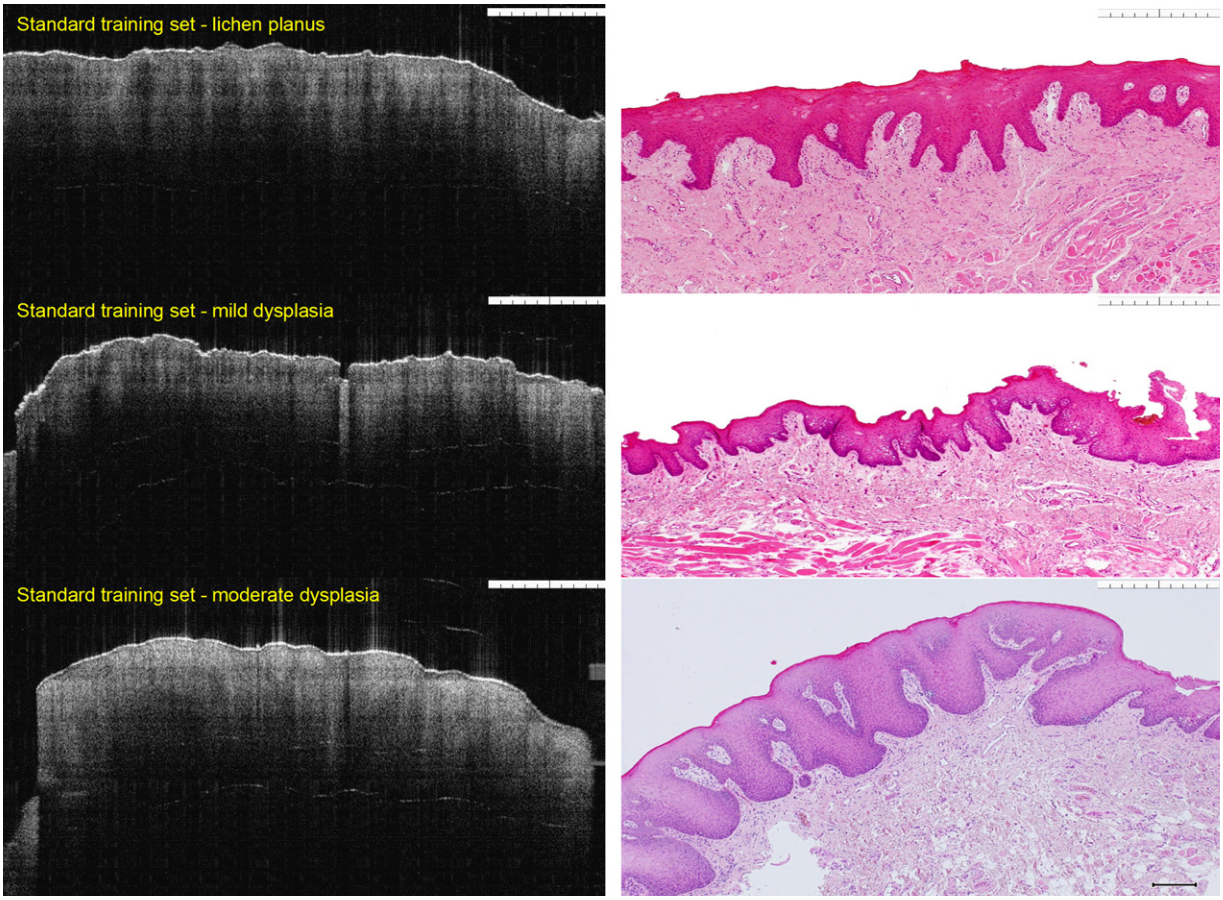

Figure 2 Data (OCT images) used in training both the clinician and the pathologist in identifying pathological processes. The images were compared to the gold standard histopathology. Here the images include lichen planus, mild and moderate dysplasias.

and excellent = 0.81—1.00). The average thickness of the keratin and epithelial layers were calculated and correlated with the corresponding pathology using Spearman correlation coefficient.

Results

Seventy two males (57.6%) and fifty three females (42.4%) participated in this clinical study; their age range was 20—88 years and 32—89 years, respectively. The median age was 58 years. Most of the biopsied lesions were located on the posterior dorsal tongue ($n = 28$), followed by buccal mucosa ($n = 20$), anterior dorsal tongue ($n = 12$), floor of mouth ($n = 11$), soft palate ($n = 11$) and vermilion border of lower lip ($n = 10$) (Table 1).

All the patients presented with clinical features suggestive of potentially malignant oral mucosal disease even those which diagnosed later as benign lesions. The majority of the lesions ($n = 65$) appeared clinically as plaques, 40 lesions were papular and 20 presented as ulcers. Lesion colour was variable: 85 (68.0%) were leucoplakic, 26 (20.8%) were erythroplakic, 13 (10.4%) were leukoerythroplakic and 1 (0.8%) was bluish (Table 1). The pathological diagnosis revealed 43 microinvasive carcinomas and 41 dysplasias (potentially malignant disorder). Benign oral lesions were less common and included 22 keratosis, 11 non-specific inflammatory reaction, 6 mucocels and 2 papillomas (Table 1).

Assessment of the epithelial structures by both the pathologist and clinician was recorded in Table 2. Each OCT image from the diseased area was accompanied with OCT image from normal epithelium of the same anatomic area. The pathologist's sensitivity and specificity was 78% and 81%, respectively. Whilst the positive predictive value (PPV) was 87% and the negative predictive value (NPP) was 70%. The accuracy of OCT was 79% (Table 3). Sensitivity and specificity for the surgeon were calculated and was 92% and 75%, respectively. PPV was 86% and NPP was 85%. The accuracy of OCT was 85% (Table 4). The collective sensitivity and specificity for both assessors was 85% and 78%, respectively. The PPV and NPV was 86.5% and 77.5%, respectively. The assessors report on the differential diagnosis is summarized in Table 5.

The average keratin and epithelial layers thickness were variable per pathological process. The mean thickness for the benign lesions was 24 μm and 338 μm, respectively. For the dysplasia (potentially malignant disorder) group the mean thickness was 23 μm and 455 μm for the keratin and epithelial layers, respectively. Carcinoma *in situ* mean thickness on the OCT image was 29 μm and 570 μm, respectively. In the presence of microinvasive carcinoma, the average

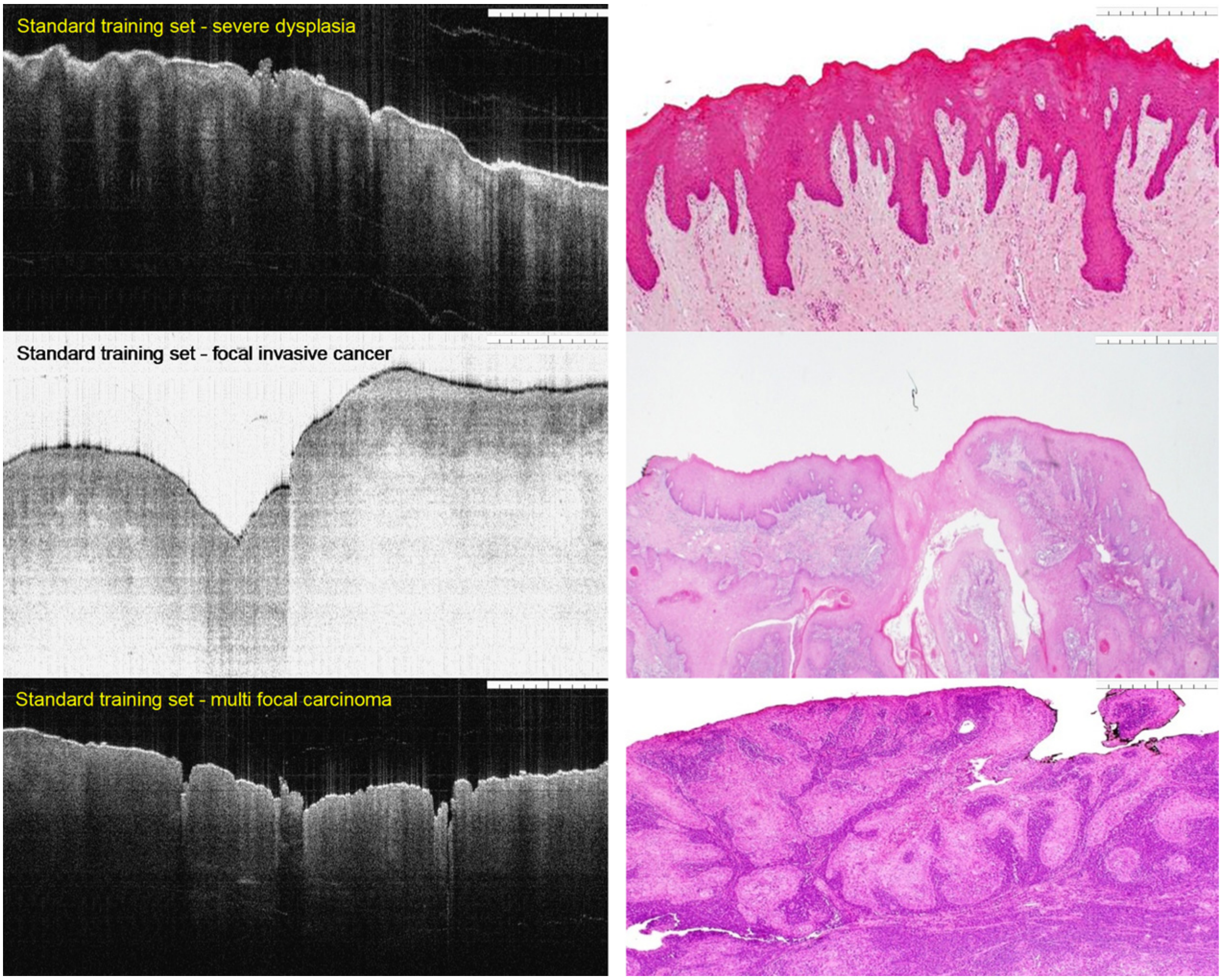

Figure 3 Data (OCT images) used in training both the clinician and the pathologist in identifying pathological processes. The images were compared to the gold standard histopathology. Here the images include severe dysplasia, focal and multi-focal carcinomas.

thickness of the epithelium was 650 μm which was higher than any of the pathological processes; while the keratin layer thickness was the lowest (19 μm) (Table 6).

No correlation has been found between the thickness of keratin cell layer and the severity of the disease. Higher correlation was found between total epithelial thickness and lesions associated with dysplasia and invasive carcinoma.

The Kappa agreement was variable for the subjective OCT images analysis. There was poor agreement on keratin layer thickness, good agreement on the epithelium layer thickness, poor agreement on the lamina proprial changes and good agreement when assessing the status of the basement membrane. Agreement on ''the need for biopsy'' was very good (Table 7).

Discussion

The diagnosis of oral cancer and or pre-cancer still requires taking surgical biopsy to study the sampled tissue histologically. Oral biopsies are usually associated with pain, bleeding and difficulties in eating and speaking [15]. Alternatively, different optical diagnostic modalities have emerged. OCT techniques for medical diagnosis have been under investigation for many years, but most of the early studies were concentrated on the use of OCT in ophthalmic applications [16].

Preliminary work has recently been presented on OCT for squamous epithelium of the upper aerodigestive tract, but the field of OCT diagnosis of oral lesions is still largely underexplored [17,18]. In our previous preliminary study, we demonstrated the feasibility of OCT to differentiate between malignant and non-malignant oral mucosal disease. A thorough understating regarding the histopathological features has been obtained and precisely identified. However, the study failed to demonstrate clear cut diagnostic information for potentially malignant disorders. The comparison between the OCT images and the H&E pathology slides was not always possible due to the tissue shrinkage and formalin effect after surgical excision [19].

The real question in this study we hoped to address concerns the value of OCT in the normal clinical setting. High-resolution optical imaging techniques such as confocal microscopy have the potential to image changes in tissue architecture and cellular morphology [20]. Reflectance imaging techniques like fluorescent spectroscopy, on the other hand, can detect local changes in scattering and absorption due to changes in the biochemical endogenous fluorophores [21].

Table 2 Assessment of the epithelial structures by both the pathologist and clinician.

Pathologist	No. (%)	Clinician	No. (%)
Keratin layer		Keratin layer	
Increased	59 (47.2)	Increased	61 (48.8)
Decreased	19 (15.2)	Decreased	31 (24.8)
No change	47 (37.6)	No change	33 (26.4)
Epithelial layer		Epithelial layer	
Increased	72 (57.6)	Increased	84 (67.2)
Decreased	3 (2.4)	Decreased	5 (4.0)
No change	50 (40.0)	No change	36 (28.8)
LP layer		LP layer	
Obvious changes	34 (27.2)	Obvious changes	26 (20.8)
No clear changes	20 (16.0)	No clear changes	20 (16.0)
No changes	71 (56.8)	No changes	79 (63.2)
BM quality		BM quality	
Excellent	23 (18.4)	Excellent	1 (0.8)
Good	50 (40.0)	Good	35 (28.0)
Adequate	32 (25.6)	Adequate	73 (58.4)
Poor	20 (16.0)	Poor	16 (12.8)
BM status		BM status	
Intact	78 (62.4)	Intact	77 (61.6)
Breach	35 (28.0)	Breach	47 (37.6)
Difficult to assess	12 (9.6)	Difficult to assess	1 (0.8)

The grading system for the assessment of oral epithelial dysplasia that was set by 'Smith and Pindborg' is partly applicable in OCT [22]. This is mainly due to the ability of the OCT to highlight only morphological changes rather than cellular. One study assumed that the increase of mitotic activity increases nuclear presence, thus cytoplasmic ratio and loss of cellular cohesiveness will collectively affect epithelium thickness [23].

The ability of OCT to precisely measure the keratin cell layer and epithelium thickness is a paramount factor. However, this study demonstrated that the thickness of the keratin cell layer alone is not informative. There is a marginal difference between all the groups where it is hard to ascertain whether the increase of the layer is part of the cancerous process or reactive to local irritation factors.

Within the same group of benign lesions, in the hyperkeratosis, there is only an increase in the thickness of the parakeratin or orthokeratin layer of the epithelium, while the other part of epithelium remains normal with or without some degree of hyperplasia. On the other hand, epithelial dysplasias may or may not show parakeratinization, which might reflect the stage of dysplasia and cellular immaturity. This means that assessment of these layers individually by OCT is fruitless.

Furthermore, oral mucocele demonstrates the lowest thickness group for the keratin cell layer (as extravasation cysts they are expected to have a thin external layer), while keratosis holds the highest. Most interestingly, the low value for the keratin cell layer was linked with invasive carcinoma. One explanation to this finding is the atrophy of the most superficial layer especially when the lesion is manifested clinically as erythroplakia and/or ulceration suggesting ischaemia of the superficial layers.

The highest epithelium thickness was found in cancerous lesions. Dysplastic lesions show fluctuation in the thickness, so it is difficult to draw a conclusion between the benign and early pre-cancer stage. Our results demonstrated that it was

Table 3 Pathological assessment. Sensitivity, 78%; specificity, 81; positive predictive value (PPV), 87%; negative predictive value (NPV), 70%; accuracy of OCT, 79%. TP: true positive; FP: false positive; TN: true negative; FN: false negative.

	Histology		
	Positive	Negative	Total
OCT			
Positive	61 (TP)	9 (FP)	70
Negative	17 (FN)	38 (TN)	55
Total	78	47	125

Table 4 Clinical sassement. Sensitivity, 92%; specificity, 75%; positive predictive value (PPV), 86%; negative predictive value (NPV), 85%; accuracy, 85%. TP: true positive; FP: false positive; TN: true negative; FN: false negative.

	Histology		
	Positive	Negative	Total
OCT			
Positive	72 (TP)	12 (FP)	84
Negative	6 (FN)	35 (TN)	41
Total	77	48	125

Table 5 Summary of the deferential diagnosis of both the pathologist and clinician when assessing the OCT images of the 125 patients.

Pathologist	No. (%)	Clinician	No. (%)
Differential 1		Differential 1	
Benign	28 (22.4)	Benign	14 (11.2)
Cyst	2 (1.6)	Cyst	7 (5.6)
Papilloma	8 (6.4)	Vesiculobollus	8 (6.4)
Hyperkeratosis	15 (12.0)	Papilloma	7 (5.6)
Lichen planus	9 (7.2)	Polyp	1 (0.8)
Dysplasia	26 (20.8)	Hyperkeratosis	16 (12.8)
Carcinoma *in situ*	23 (18.4)	Lichen planus	16 (12.8)
SCC	14 (11.2)	Dysplasia	25 (20.0)
Differential 2		Carcinoma *in situ*	26 (20.8)
Cyst	4 (3.2)	SCC	5 (4.0)
Vesiculobollus	2 (1.6)	Differential 2	
Papilloma	3 (2.4)	Cyst	2 (1.6)
Polyp	16 (12.8)	Vesiculobollus	3 (2.4)
Hyperkeratosis	12 (9.6)	Papilloma	5 (4.0)
Lichen planus	1 (0.8)	Polyp	7 (5.6)
Dysplasia	22 (17.6)	Hyperkeratosis	15 (12.0)
Carcinoma *in situ*	14 (11.2)	Lichen planus	4 (3.2)
SCC	27 (21.6)	Dysplasia	32 (25.6)
None	24 (19.2)	Carcinoma *in situ*	23 (18.4)
Differential 3		SCC	27 (21.6)
Vesiculobollus	3 (2.4)	None	7 (5.6)
Polyp	3 (2.4)	Differential 3	
Hyperkeratosis	15 (12.0)	Vesiculobollus	1 (0.8)
Dysplasia	2 (1.6)	Papilloma	1 (0.8)
Carcinoma *in situ*	11 (8.8)	Polyp	3 (2.4)
SCC	5 (4.0)	Hyperkeratosis	7 (5.6)
None	86 (68.8)	Lichen planus	1 (0.8)
Need for biopsy	70 (56.0)	Dysplasia	14 (11.2)
		Carcinoma *in situ*	12 (9.6)
		SCC	24 (19.2)
		None	62 (49.6)
		Need for biopsy	84 (67.2)

possible to distinguish healthy/benign lesions from advanced dysplastic lesions. We achieved excellent results in discriminating cancerous tumours from healthy tissue. Interestingly, this is based on good agreement between observers on basement membrane status.

Our results confirm and expand the findings of previous studies. Wilder-Smith et al. [24] analysed 50 patients with different oral lesions obtained *in vivo*. The assessment of the sensitivity and specificity of diagnostic algorithm was based on the criteria established by MacDonald [25], which score epithelial dysplasia in hamster cheek pouch according to the criteria based on human oral mucosa. We faced difficulties when differentiating between different grades of oral dysplasia using the classification system of the cellular component of Smith and Pindborg.

Our sensitivity and specificity in general is lower than expected. This may be attributed to two factors. First: we used the *ex vivo* samples which lack capillary fluid perfusion which eventually affect optical scattering properties and image resolution. Effects of ischaemia, such as from cell lysis and necrosis, alter tissue micro-structure and therefore can affect optical scattering and imaging contrast. Second: the differences were almost certainly related to the difference in the methodologies of the studies for scoring the sensitivity and specificity.

Fomina et al. [26] investigated the capability of *in vivo* OCT for the diagnosis of oral lesions. In comparison to our results, the specificity of this study was much higher with very good inter-observer agreement. We assumed that the '*in vivo*' application of OCT helps clinicians to be more sensitive and specific. This speculation comes from the fact that the contralateral anatomical side helps to control the benchmark reference point. In terms of specificity, the former study reported 83% which is within the scale of our result.

The unanticipated result of the present study was the poor agreement on the changes in the keratin cell layer. The reasons for this were unclear. However, the training exercise was robust and placed greater emphasis on reducing discrepancy. The degree of variation to interpret this parameter does not influence the accuracy.

Another important outcome of the statistical analysis using epithelium thickness is the ability of OCT to separate benign lesions from all other lesion types despite having

Table 6 The average keratin layer (KL) and epithelial layer (EL) thickness per pathological process.

Pathology	Range	Minimum	Maximum	Mean
Benign				
KL thickness	34 μm	8 μm	42 μm	24 μm
EL thickness	405 μm	175 μm	580 μm	338 μm
Mucocele				
KL thickness	4 μm	8 μm	12 μm	10 μm
EL thickness	55 μm	175 μm	230 μm	195 μm
Keratosis and palilloma				
KL thickness	27 μm	15 μm	42 μm	28 μm
EL thickness	370 μm	197 μm	567 μm	367 μm
Non-specific				
KL thickness	31 μm	10 μm	41 μm	23 μm
EL thickness	346 μm	234 μm	580 μm	387 μm
Dysplasia (all)				
KL thickness	25 μm	15 μm	40 μm	23 μm
EL thickness	606 μm	170 μm	776 μm	455 μm
Mild dysplasia				
KL thickness	10 μm	16 μm	26 μm	20 μm
EL thickness	262 μm	170 μm	432 μm	297 μm
Moderate dysplasia				
KL thickness	18 μm	15 μm	33 μm	20 μm
EL thickness	361 μm	232 μm	593 μm	453 μm
Severe dysplasia				
KL thickness	19 μm	17 μm	36 μm	25 μm
EL thickness	542 μm	234 μm	776 μm	484 μm
Carcinoma *in situ*				
KL thickness	11 μm	23 μm	34 μm	29 μm
EL thickness	41 μm	549 μm	590 μm	570 μm
SCC				
KL thickness	32 μm	5 μm	37 μm	19 μm
EL thickness	589 μm	385 μm	974 μm	650 μm

the same or similar outward appearance. Many optical techniques are capable of discriminating normal from malignant oral mucosa with a high degree of sensitivity and specificity, but discrimination of benign lesions from pre-cancer is more challenging [27].

Kraft et al. found the values of epithelium thickness in laryngeal mucosa to be very informative. The mean value for the normal mucosa was 147 μm, benign lesion 241 μm (244 μm for polyp, 248 μm for papilloma), mild dysplasia (258 μm), 301 μm for moderate dysplasia), severe dysplasia (44 μm), carcinoma *in situ* (496 μm) and micro-invasive carcinoma (974 μm) [23].

Table 7 Kappa agreement between the pathologist and the clinicians when assessing the OCT images for the 125 patients.

Assessment of pathologist *vs.* clinician	Kappa
Biopsy	0.720
Keratin layer	0.279
Epithelial layer	0.452
Lamina properia	0.306
Basement membrane quality	0.590
Basement membrane status	0.656

In order to discriminate between benign or premalignant oral lesions from frank malignant tumuor, the basement membrane has been used as a key landmark. The basement membrane aided the scorers to decide whether the lesions are cancerous or not. Other parameters helped to differentiate pre-cancer from benign included epithelial thickness and other architectural changes.

Despite the excellent sensitivity, specificity and good agreement on the basement membrane status, we have found that there was only moderate agreement on the basement membrane quality. This means that the basement membrane, as a parameter, held great subjectivity. However, it seems that there was an issue regarding its quality. One may conjecture that the most likely causes of the good agreement, rather than very good or excellent, may be due to the shadows produced from partial tissue thickness which is different from the real clinical situation.

Our study suggested that good agreement can be achieved in the assessment of the epithelium thickness. Our overall kappa (0.452) was slightly lower than the result from the basement membrane. The intra-observer agreement was similar for lamina propria and keratin cell layer. It is reassuring to observe good agreement for ''the need for biopsy'' between the observers, despite the discrepancy in the sensitivity and specificity.

Poor agreement was seen in the status of the lamina propria is concerning. A number of factors have been suggested, including lack of tissue perfusion that may alter light scattering or decrease light penetration. It is also possible that about one third of the suspicious lesions were originally SCCs, which may bewilder the reader regarding obscured status of the lamina properia in some areas.

More research will be required to characterize the normal thickness of oral mucosa together with reference ranges for malignant and benign conditions. One can then clearly envision and standardize the normal average. This is the next stage of our work. An assessment of the epithelium thickness in the dysplastic lesions has good reliability in our study and is reliable in the diagnostic grading of oral potentially malignant disorders.

One limitation of this study was the '*ex vivo*' protocol. The authors expect that the future '*in vivo*' images will be far superior in term of quality and resolution. This may potentially improve the final sensitivity and specificity. Another limitation which should be addressed in future research is the evaluation of the accuracy of OCT to differentiate between benign lesions, dysplastic and cancerous lesions, as well as differentiating dysplasia from invasive carcinoma.

Conclusion

This study has shown some inconsistent results, especially with regards to the deceptive change in the histological layers on optical analysis when compared to conventional biopsy. Furthermore, OCT image analysis was unique and associated with a wide range of variability when interpreting its parameters.

The epithelium thickness and the status of the basement membrane play a key role for OCT interrogation of tissues. The keratin cell layers provide little diagnostic information alone. However, combining the clinical information of epithelium thickness and keratin cell layer may provide valuable diagnostic information.

Competing interests

Mr Colin Hopper is an advisory board member at Michelson Diagnostics, Kent, UK.

Dr Gordon McKenzie is a medical applications director at Michelson Diagnostics, Kent, UK.

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